Whereas gRPC works best with server-to-server communication and nonbrowser clients, GraphQL is best used to provide evolvable APIs to mobile and single-page application (SPA) apps. It has become popular among frontend developers, as it can reduce the friction involved in deploying and using new APIs. For details, I recommend *GraphQL in Action*, by Samer Buna (Manning, 2021).

Despite the benefits and improvements gRPC and GraphQL can bring, RESTful HTTP services are here to stay for the foreseeable future, so it's worth making sure that you understand how to use them with `HttpClient`.

In .NET we use the `HttpClient` class for calling HTTP APIs. You can use it to make HTTP calls to APIs, providing all the headers and body to send in a request, and reading the response headers and data you get back. Unfortunately, it's hard to use correctly, and even when you do, it has limitations.

The source of the difficulty with `HttpClient` stems partly from the fact that it implements the `IDisposable` interface. In general, when you use a class that implements `IDisposable`, you should wrap the class with a `using` statement whenever you create a new instance to ensure that unmanaged resources used by the type are cleaned up when the class is removed, as in this example:

```
using (var myInstance = new MyDisposableClass())
{
    // use myInstance
}
```

TIP C# also includes a simplified version of the using statement called a `using` *declaration*, which omits the curly braces, as shown in listing 33.1. You can read more about the syntax at http://mng.bz/nW12.

That might lead you to think that the correct way to create an `HttpClient` is shown in listing 33.1. This listing shows a simple